

COMPUTER NETWORKING I

Project Team

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EVALUATION OF TECHNOLOGICAL NETWORK INFRASTRUCTURE AND DIGITAL
SERVICE COSTS AT DISTRITAL UNIVERSITY

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CURRICULAR PROJECT IN SYSTEMS ENGINEERING

COMPUTER NETWORKING I

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Part I

CONTEXT OF THE PROBLEM

1 INTRODUCTION

The Universidad Distrital Francisco José de Caldas relies on a robust technological infrastructure to support its academic, administrative, and research activities. However, as the institution continues to grow, so does the demand for a high-performance network capable of ensuring seamless connectivity across all faculties and campuses.

This project aims to analyze and estimate the costs associated with deploying a comprehensive network infrastructure that can support the entire university. The study will focus on defining the necessary network requirements, including bandwidth, network topology, security measures, and scalability to accommodate future technological advancements. Additionally, the project will evaluate the number of users, services that must be supported, and the type of hardware and software needed to ensure optimal network performance.

To provide a well-rounded assessment, we will conduct two different cost estimations. First, we will create an ideal estimation, considering all necessary resources without financial constraints, ensuring maximum performance and security. Then, we will develop a cost-efficient estimation, optimizing expenses by identifying areas where costs can be reduced while maintaining an efficient and functional infrastructure. This comparative approach will help determine the most suitable investment strategies for the university's technological expansion.

By assessing both financial and technical feasibility, this project aims to provide valuable insights for decision-making in the modernization and expansion of the institution's digital services. The outcome will contribute to an improved educational experience, streamlined administrative operations, and enhanced research capabilities through a secure and high-speed network environment.

2 OBJECTIVES

GENERAL OBJECTIVE

To analyze, design, and estimate the costs of a network infrastructure capable of supporting all faculties, campuses, and digital services of Universidad Distrital Francisco José de Caldas, ensuring high performance, security, scalability, and accessibility. The project will provide two cost estimations: one considering all necessary resources without financial constraints and another focused on cost optimization while maintaining an efficient and functional network.

SPECIFIC OBJECTIVES

Specific objectives for this project are:

- Define the functional and technical requirements of the university-wide network infrastructure, including the number of users, necessary services, bandwidth, security measures, and hardware/software specifications.
- Comparing two cost estimations applicated in the same context: an ideal estimation and a cost-optimized estimation.
- Propose an implementation and management strategy for the network infrastructure, ensuring long-term reliability, security, and adaptability to future technological advancements.

3 PROBLEM DESCRIPTION

As Universidad Distrital Francisco José de Caldas continues to grow, so does the demand for a robust, scalable, and secure network infrastructure capable of supporting its academic, administrative, and research activities. The university serves approximately 26,000 students, 3,000 faculty and administrative staff, and thousands of visitors and alumni, with an estimated total of 30,000 to 35,000 potential users connected simultaneously. Each user may connect multiple devices (laptops, smartphones, tablets), resulting in an estimated demand for at least 50,000 concurrent connections across all campuses.

Currently, the university operates a complex digital ecosystem, which includes:

- A main institutional website (www.udistrital.edu.co)
- Around 50 faculty and administrative websites
- A Virtual Learning Environment (Aulas Virtuales)
- The Academic Management System (SGA)
- A digital repository for research and publications
- More than 20 additional online platforms and services, including email, library access, financial management, research portals, and internal communication tools

However, the existing network infrastructure may not be fully optimized to handle this increasing demand. Insufficient bandwidth, security vulnerabilities, network congestion, and scalability limitations pose significant challenges. Additionally, a lack of redundancy and disaster recovery mechanisms may threaten the reliability of university services, potentially affecting thousands of students and staff.

Designing and deploying a network at such a large scale involves significant financial and technical challenges. The university must balance performance, security, and scalability while ensuring a cost-effective investment in technology.

To address this issue, the project will explore two different cost estimations: an ideal estimation that considers all necessary resources without financial constraints, ensuring maximum performance and security, and a cost-efficient estimation that optimizes expenses by identifying areas where costs can be reduced while maintaining a reliable and scalable network.

Part II

REQUIREMENTS

4 Functional Requirements

4.1 provide services to all user groups during typical service usage

The main users of the university network can be classified as:

- **Students:** Undergraduate and postgraduate students actively using online services.
- **Faculty and Administrative Staff:** Professors, researchers, and administrative employees.
- **Visitors and Alumni:** External users, guest lecturers, event attendees, and alumni accessing online services.
- **IoT and Connected Devices:** Smartboards, security cameras, lab equipment, and other devices that generate traffic.

In the following table, it will be approximated the quantity of users of the network and the group they belong to:

User Group	Stimted Population	Average Devices per User	Stimted Connections
students	29.102	2 (Laptop and Smartphone)	52.000
Faculty and Staff	3.000	2 (Laptop and Smartphone)	6.000
Visitors	500-1000	1.5	3.000-6.000
IoT and institutional Devices	5.000	1	5.000
Total	37.102 - 39102	66.000 - 69.000 devices	66.000-69.000 conexions

Figure 1: Table of users in the Distrital University

The services offered by the university are:

Service	Purpose	estimated AP's
Academic Management System (SGA)	Enrollment, schedules, grades	30.000+ students
Virtual Learning (Aulas Virtuales)	Online classes, assignments	20.000+ users
Institutional Email	Official communication	30.000+ active accounts
Library & Repositories	Digital books, research papers	10.000+ queries
Research & Computing Labs	Simulations, cloud computing	1000+ daily
Administrative & HR Systems	Payroll, hiring, internal docs	3.000+ users
Student Services	Well-being, financial aid, medical	10+ daily
Public Website	General info, faculty pages	1600+ daily

Figure 2: Table of services of the Distrital University

Each service requires a database, application, and storage servers, plus redundancy to ensure high availability, but not all users will be connected simultaneously, so we estimate a peak concurrent usage of about 35-40% (even less) of total connections in a normal situation, so 3 or 4 servers would be more than enough to handle university's services.

About the storage capacity, assuming the university operates at 40% peak concurrency, the estimated daily bandwidth usage is:

Service	Estimated Daily Bandwidth Usage (TB)
Academic Management System (SGA)	0.6 TB/day
Virtual Learning (Aulas Virtuales)	1.6 TB/day
Institutional Email	0.06 TB/day
Library & Repositories	0.4 TB/day
Research & Computing Labs	1.0 TB/day
Administrative & HR Systems	0.06 TB/day
Public Website	0.4 TB/day

Figure 3: Estimation of the bandwidth usage for the Distrital University per day

And in a year, the estimation of the storage usage is:

Service	Estimated Storage Usage (TB)
Academic Management System (SGA)	10 TB/year
Virtual Learning (Aulas Virtuales)	10 TB/year
Institutional Email	5 TB/year
Library & Repositories	10 TB/year
Research & Computing Labs	5 TB/year
Administrative & HR Systems	5 TB/year
Public Website	10 TB/year

Figure 4: Estimation of the storage usage for the Distrital University in a year

4.2 provide services to all user groups during peak service usage

The situations where there is significantly higher-than-usual traffic on Universidad Distrital’s network occur during the publication of class schedules in the academic management system, as all students want to check their enrolled courses. However, by far, the highest traffic happens during the academic course registration period.

The enrollment process takes place over an entire week, where each day is assigned to a specific faculty, allowing its students to register for courses. This continues until all faculties have had their turn, followed by two additional days when all students can enroll.

Since there are no official figures on how many students are enrolled in undergraduate and postgraduate programs for the year 2024, and the last published data was from 2015—showing a proportion of 22,922 undergraduate and 1,388 postgraduate students, which corresponds to 94.29% undergraduate and 5.71% postgraduate—we assume that these percentages have remained the same or at least very similar. Applying these proportions to the total of 29,102 students enrolled until the third academic

term of 2023, we estimate that there are approximately 27,450 undergraduate students and 1,660 post-graduate students.

Considering the number of undergraduate students, who are the ones required to enroll in courses and also those who rush to check their schedules when they are released, we will classify them by faculty.

For this purpose, we searched for the exact number of students in each faculty but only found precise figures for the Faculty of Engineering and the Faculty of Science and Education, with 5,243 and 7,028 undergraduate students, respectively.

Therefore, for the remaining five faculties, we took the estimated total number of undergraduate students (27,450), subtracted the sum of 5,243 and 7,028, and divided the result by four (taking into account that the health sciences faculty has no students yet) to obtain an estimate of the number of undergraduate students per faculty, which is approximately 3,794 students each.

Faculty	Stimated Population
Sciences and Education	7.028
Engineering	5.243
Faculty of Environmental	3795
Technology Faculty	3.795
Faculty of Arts ASAB	3.795
Faculty of Mathematics	3.795
TOTAL	27.447

Figure 5: Distribution of students in the Distrital University by faculty.

In the worst-case scenario, we can assume that on the enrollment day, 95% of the students from the Faculty of Science and Education attempt to access the enrollment service. This means that, at peak usage, we would have approximately 6,677 users trying to connect.

If we assume that each user attempts to do so using two devices in order to secure the academic slots they need, the system would receive approximately 13,354 service requests. The necessary bandwidth to support 13,354 concurrent visits, passes from 0.6TB to is needed for a normal day with 300 visits, is approximately

4.3 Wifi Full Coverage

Every faculty must provide complete WiFi coverage to accommodate students, faculty, and administrative staff.

- The network must support high-density environments, ensuring that a large number of devices can connect simultaneously without performance degradation.
- WiFi 6 (or newer) technology should be implemented to enhance connectivity, reduce congestion, and optimize efficiency.
- Access points (APs) should be strategically placed to avoid dead zones and guarantee seamless connectivity across classrooms, laboratories, libraries, and common areas.
- Each faculty's WiFi must integrate with the main university network, ensuring secure and controlled access to academic and institutional services.

- Guest and visitor access should be managed separately from student and faculty networks to maintain security and optimize bandwidth allocation.

Based on the estimated number of students, administrative staff (estimated 200 per faculty), and visitors per faculty, here are the approximate total users per faculty, considering each faculty has its own independent network:

- Faculty of Engineering: 5,543 users
- Faculty of Science and Education: 7,328 users
- Each of the other four faculties (estimated): 4,136 users

Each faculty will have its own dedicated WiFi network, meaning we need 7 independent faculty networks plus an additional guest network for visitors, events, and external users.

Total WiFi Networks Needed: 7

- 1 per faculty (7 total)
- 1 general guest network

Each faculty's network should be divided into VLANs to separate students, faculty, research labs, and administrative staff, ensuring security and optimized traffic flow.

We calculate bandwidth requirements based on:

- Simultaneous connections: Assume 70% of users are connected at peak usage.
- Bandwidth demand per user (based on typical university usage patterns).
 1. Light users (email, browsing) → 2 Mbps per user
 2. Moderate users (video streaming, downloads) → 5 Mbps per user
 3. Heavy users (research labs, cloud computing, online classes) → 10 Mbps per user
 4. Extreme users (engineering/software labs, VR, simulations) → 15 Mbps per user

Total estimated bandwidth per faculty (assuming a weighted distribution of user types):

Faculty	Total Users	Stimated Peak Users (70%)	Required Bandwidth (Gbps)
Sciences and Education	5.543	5.130	30 Gbps
Engineering	7.328	3.880	25 Gbps
Faculty of Environmental	4.136	2.895	20 Gbps
Technology Faculty	4.136	2.895	20 Gbps
Faculty of Arts ASAB	4.136	2.895	20 Gbps
Faculty of Mathematics	4.136	2.895	20 Gbps
TOTAL	33.775	23.895	150 Gbps

Figure 6: Distribution of students in the Distrital University by faculty.

4.3.1 Access Point (AP) Density Estimate

Assuming one AP covers 50-70 simultaneous users, we calculate:

Faculty	Total Users	Stimated Peak Users (70%)	estimated AP's
Sciences and Education	7.028	5.130	80
Engineering	5.243	3.880	60
Faculty of Environmental	3795	2.895	45
Technology Faculty	3.795	2.895	45
Faculty of Arts ASAB	3.795	2.895	45
Faculty of Mathematics	3.795	2.895	45
TOTAL	27.447	23.895	320

Figure 7: Distribution of students in the Distrital University by faculty.

5 Non Functional Requirements

5.1 Security concerns

5.1.1 Physical infrastructure

For the on-premise part of the data center, physical space will be necessary to store all the servers and devices needed:

- **HVAC:** heating, ventilation and air conditioning is essential for the data center; the amount of heat they emanate is dangerous for the devices. So at first we gotta calculate the heat load; every device of the data center, such as servers, storage devices, electrical devices, and even the operators, has to be taken into account to calculate this heat load and the future devices that can be added.

To ensure proper temperature and humidity control, the data center will require:

1. CRAC (Computer Room Air Conditioners): 3-5 CRAC units, depending on the heat load calculation.
2. Airflow Ventilation System: Filtered air intake/exhaust to prevent dust and contaminants from entering.
3. Precision Cooling Units: For temperature-sensitive equipment.
4. Humidity Monitoring Sensors: 5-10 smart sensors across the facility.

CRAC (Computer Room Air Conditioners) will be used in this design to control temperatures when it is needed, because also other cooling systems will be implemented such as airflow ventilations to exterior (properly filtered to avoid dust and dangerous agents entering the data center).

- **Physical security:** Multiple factors has to be take on sighth in this aspect, unauthorized access will be controled by multiple security implementations, fingerprint secured doors will be used, also a security and surveillance third party service will be used to keep guards on multiple places looking for the security of the place, a CCTV (Closed Circuit Television) with strategically placed cameras will be used so it's easier to keep a good sighth of every corner of the data center; adding a mantrap is also a good option but we think this can be out of budget because it is really expensive.

To prevent unauthorized access and protect physical assets, the following will be required:

1. Fingerprint-secured doors: 2-4 biometric doors for controlled entry.
 2. Security Guards: Contract with a third-party security company (24/7 monitoring).
 3. CCTV System: 12-20 high-resolution security cameras with night vision and motion detection.
 4. Mantrap (Optional): Expensive but effective dual-door security system.
 5. Environmental Sensors: Leak detection, dust monitoring, humidity temperature sensors.
- **Electrical safety protocol:** As every device works with an electrical power source this is a really imporant part of the data center, the power sources will have a UPS (Uninterrupted Power Source) system and backup generators, that will provide enough power in case that a failure on the main electricity system happends. Also circuit breakers, fuses and a proper electrical isolation will help to prevent short circuits, overloads and fire.

A desing of the cable management on the data center will be made so it stays organized and maintainable, apart from preventing damage on them.

For this purpose is necessary:

1. UPS (Uninterruptible Power Supply) Systems: 3-5 high-capacity UPS units for redundancy.
2. Electrical Circuit Breakers and Fuses: Installed across racks to prevent overloads.
3. Organized Cable Management System: Structured cabling design for safety and maintenance.

5.1.2 Logical security

Cyber security is a very important part of the data center:

- Firewall: this acts like a barrier against cyber attacks, it blocks unauthorized and malicious traffic and will be used as one of the security tools for the data center.
- Compliance: a normative documentation for users and workers to avoid risks will be created, using multiple standards such as SOC 2 for data centers data security, TIA-942 for physical security, LEED for friendly environmental data centers, OSHA for safe working, Habeas Data Law for data protection.
- Authentication: a multifactor authentication to avoid unauthorized accesses of the data, we will put policies to have strong passwords and role based access control. We will implement SSH, RPD access, keyrotation systems, we will use VLANs and VPN to isolate the network so it becomes safer.
- Antiviruses and DDoS protection: using antiviruses in endpoints is a need, it will be used on servers, workstations, etc. so viruses won't spread. In the side of the DDoS protection a perimeter protection will work because it detects and blocks malicious traffic before it reaches the data center, also limiting the rate of requests also monitoring the network will provide a better security, also all of this can be made by the AWS Shield protection.

Part III

TECHNICAL LIMITATIONS

the data center may face various technical limitations and constraints. Some of these challenges can be mitigated by the university, while others depend on external factors beyond institutional control. Below are the most significant barriers that may impact performance, scalability, and security:

- **Storage Constraints:** The estimated storage capacity of the data center may not be sufficient to meet the growing demands of users. High data traffic and storage-intensive applications, such as research repositories, cloud-based learning platforms, and administrative databases, could surpass the expected 40% peak concurrency model. While scalability is a viable solution, expanding storage would require:
 1. Additional physical space for new storage units.
 2. Enhanced cooling systems to manage increased heat output.
 3. Higher energy consumption, leading to increased operational costs.
- **Energy Consumption and Power Stability:** A data center operating 24/7 requires a significant amount of energy, which may put strain on the local power grid if the energy provider is not equipped to handle such a high load. While this issue is partially addressed by the electrical safety protocol (which includes UPS systems, backup generators, and load balancing), it adds complexity to data center maintenance and infrastructure control.
- **Internet Service Provider (ISP) Limitations:** The university's data center and faculties require bandwidth ranging from 20 Gbps to 30 Gbps to ensure seamless connectivity and low-latency performance. However, the available ISP infrastructure may not always meet these demands, leading to slower network speeds, higher latency, and limited redundancy. Since ISP capacity is an external factor, this limitation cannot be controlled by the university, making it a critical dependency for the success of the project.
- Even with high-speed fiber-optic connections and network redundancy, peak usage events—such as course enrollment periods—can cause:
 1. Severe slowdowns and degraded performance.
 2. Increased latency and packet loss.
 3. Server failures due to excessive concurrent requests.

If the network is not properly optimized, these issues could impact academic operations and disrupt essential university services.

- **Wireless Interference and Coverage Challenges:** the deployment of 45 to 80 WiFi access points per faculty improves wireless coverage and accessibility, but also introduces potential technical issues:
 1. Signal interference between overlapping APs, reducing efficiency.
 2. Dead zones if APs are not strategically placed.
 3. Overloaded access points in high-density areas (e.g., libraries, auditoriums).

To minimize these risks, advanced frequency planning (WiFi 6/6E) and dynamic bandwidth allocation will be required.

- **Cybersecurity Threats (like DDoS and Cyberattacks)**

Due to its large user base and open-access environment, a university data center is highly vulnerable to cyber threats. In fact, the university has previously experienced DDoS attacks, which resulted in:

1. Severe disruptions to network services.
2. The need for complete infrastructure restoration.
3. Financial and operational setbacks.

To mitigate such risks, the data center must implement:

1. Advanced firewalls and IDS/IPS systems.
 2. Perimeter-based DDoS protection (e.g., AWS Shield, Cloudflare).
 3. Real-time traffic monitoring and anomaly detection.
- Managing secure authentication for 25,000+ active users is a significant challenge. While role-based access control (RBAC) and multi-factor authentication (MFA) enhance security, they also introduce:
 1. Usability difficulties, requiring user adaptation.
 2. Increased administrative overhead for IT staff.
 3. Potential delays in service access due to authentication verification.

Balancing security with ease of use remains a critical design consideration.

- The data center is subject to legal and regulatory constraints, including:
 1. Habeas Data Law – Ensuring data protection and user privacy.
 2. ISO 27001 and TIA-942 – Security and infrastructure standards.
 3. OSHA and LEED – Safety and environmental compliance.

These regulations increase system complexity and require continuous monitoring to ensure compliance with evolving legal frameworks.

- If operating systems, applications, and security protocols are not regularly updated, the system becomes vulnerable to:
 1. Software bugs leading to system instability.
 2. Incompatibility issues with modern security solutions.
- Although the estimated budget is \$2.108M USD, maintaining such an infrastructure is expensive in the long run. If government funding is reduced or restricted, some aspects of the project may have to be scaled down, impacting:
 1. The scope of cybersecurity protections.
 2. Storage and processing capabilities.
 3. Network expansion and redundancy.

Long-term financial sustainability must be carefully planned to ensure continued data center performance and security.

Part IV

TECHNICAL CONSIDERATIONS

6 AWS Architecture for University Web Applications

6.1 Overview

The university's web application infrastructure will be built on AWS App Runner, a fully managed service that makes it easy to deploy web applications and APIs at scale. This architecture will support various university services, including academic management systems, virtual learning environments, and institutional repositories.

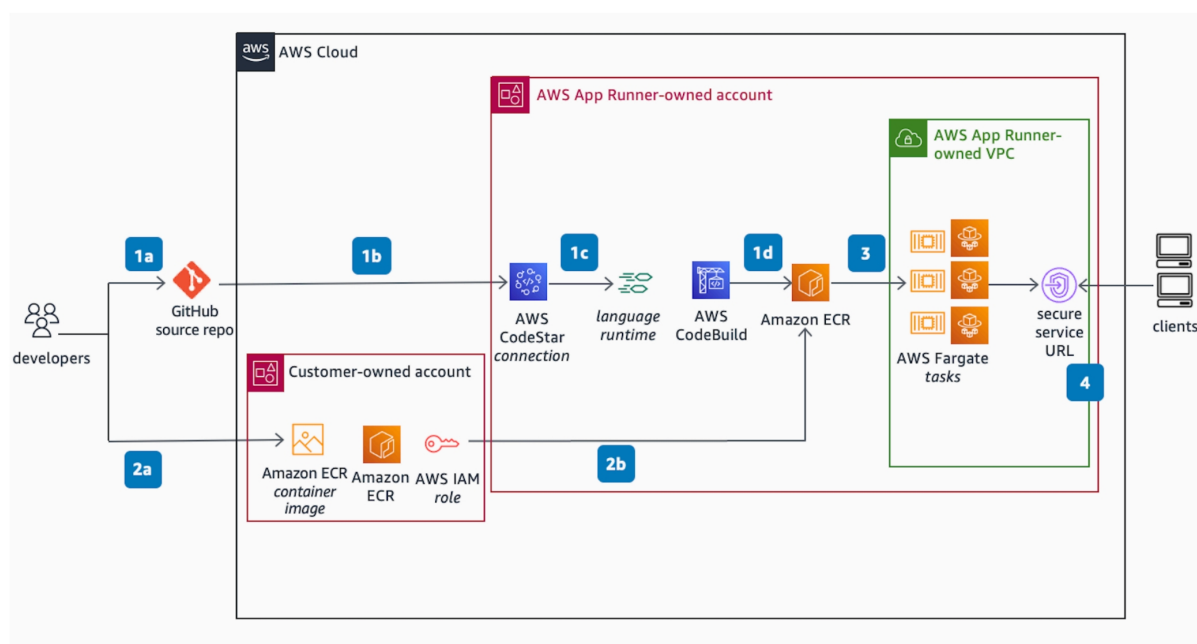


Figure 8: AWS architecture for university's web applications

The deployment workflow will consist of two main components: a source management system using GitHub repositories where developers will store and version their code, and a container management system within a customer-owned AWS account that will handle the container images and security roles. The build and deployment process will use AWS CodeStar for GitHub integration and AWS CodeBuild for creating deployment packages, all of which will be executed in an AWS App Runner-owned VPC using Fargate tasks.

The infrastructure will implement comprehensive security measures through account separation, IAM roles and secure VPC configurations, while supporting automatic scaling during peak periods such as course registration. This is achieved by load balancing across multiple Fargate tasks and elastic on-demand container provisioning to ensure optimal performance during high traffic periods.

7 "Aulas Virtuales" (Moodle) Architecture

7.1 Overview

The system will be implemented using Moodle, a highly available, scalable and elastic Learning Management System (LMS) deployed on AWS infrastructure. The system will be designed to support the University's entire student body of approximately 29,000 students, with particular attention to peak usage periods during academic activities.

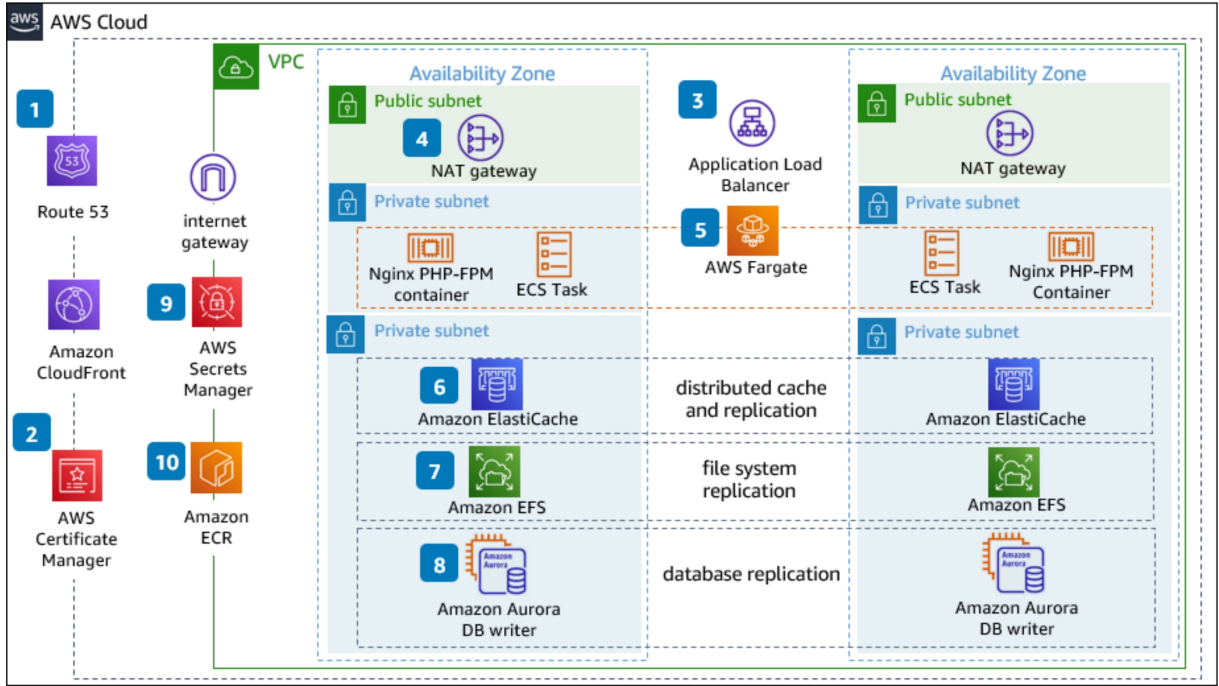


Figure 9: High Availability Moodle Architecture on AWS

The infrastructure will utilise Amazon Route 53 and CloudFront for efficient content delivery, directing students to the closest geographic locations to minimise latency. Security is managed by AWS Certificate Manager for SSL/TLS certificates and AWS Secrets Manager for sensitive data protection, while an application load balancer handles HTTPS traffic distribution across multiple availability zones. The core application will run on AWS Fargate using containerised Moodle instances, with container images stored in Amazon ECR.

The data layer is built on a robust foundation of Amazon ElastiCache for distributed caching, Amazon EFS for shared file storage, and Amazon Aurora for the primary database. This multi-tiered architecture will ensure high availability through automated database replication, distributed caching and file system replication across availability zones. The entire system will be designed to automatically scale during peak academic periods, such as exam weeks and course registration periods, and maintain optimal performance under varying loads.

Part V

BUDGET

8 Fully Coverage Budget

The planned budget for this project, acquiring the necessary devices to fully cover the requirements, and without skimping on expenses, is as follows.

ITEM	QUANTITY	UNITARY PRICE (USD)	TOTAL (USD)
SERVERS			
Blade Servers Chassis	4	15000 USD	60.000 USD
Blade Servers	20	5000 USD	100.000 USD
TOTAL: 160.000 USD			
STORAGE			
NAS System (500 TB)	1	150.000 USD	150.000 USD
TOTAL: 150.000 USD			
NETWORKING			
Core Switches	3	50.000 USD	150.000 USD
Distribution Switches	10	150.00 USD	225.000 USD
Access Points	320	230 USD	73.600 USD
TOTAL: 448.600 USD			
SECURITY			
advanced Firewalls	2	50.000 USD	100.000 USD
IDS/IPS systems	1	50.000 USD	100.000 USD
TOTAL: 200.000 USD			
INFRASTRUCTURE			
Power Distribution	1	300.000 USD	100.000 USD
Cooling System	1	250.000 USD	100.000 USD
Racks and Cabling	1	100.000 USD	100.000 USD
TOTAL: 650.000 USD			
PROFESSIONAL SERVICES			
Design and Distribution	1	500.000 USD	500.000 USD
TOTAL: 500.000 USD			
TOTAL : 2'108.600			

Figure 10: Estimation of the costs for the complete data-set.

- **Servers (\$160,000)**
 1. **Description:** Includes 4 blade chassis and 20 blade servers to support university applications, databases, and virtualization.
 2. **Justification:** Ensures high-performance computing for services like the SGA system, virtual learning platforms, and research computing. The modular design allows future scalability.
- **Storage (\$150,000)**
 1. **Description:** A 500 TB NAS system for storing academic records, research data, and university documents.
 2. **Justification:** Provides centralized, high-availability storage with redundancy (RAID, backups) to handle continuous data growth (12-18 TB/year) while ensuring long-term scalability.
- **Networking (\$500,000)**
 1. **Description:** Includes 2 core switches, 10 distribution switches, and 230 WiFi 6 access points to connect all university faculties.
 2. **Justification:** Ensures a high-speed, reliable network that supports 50,000+ devices. The WiFi 6 APs provide low-latency, high-density connectivity for students and staff.
- **Security (\$150,000)**
 1. **Description:** Investment in advanced firewalls and IDS/IPS systems for cybersecurity and network protection.
 2. **Justification:** Prevents cyber threats like DDoS attacks, unauthorized access, and data breaches. Ensures compliance with security standards (SOC 2, ISO 27001, TIA-942).
- **Infrastructure (\$650,000)**
 1. **Description:** AWS cloud services for hosting, backups, and scalability.
 2. **Justification:** Enables disaster recovery, remote learning, and secure backups. Provides elastic resources for handling peak loads (e.g., enrollment periods).
- **Infrastructure (\$650,000)**
 1. **Description:** Includes power distribution, cooling systems (CRAC units), and structured cabling for data center operations.
 2. **Justification:** Guarantees power redundancy (UPS, backup generators) and efficient cooling to prevent overheating. Critical for 24/7 uptime of university services.
- **Professional Services (\$500,000)**
 1. **Description:** Covers system design, implementation, and long-term maintenance.
 2. **Justification:** Ensures proper deployment, optimization, and monitoring of IT infrastructure. Reduces future downtime and maintenance costs.
- **Total Budget: \$2,360,000 USD**
 1. Designed to ensure scalability, security, and high availability for all university services.
 2. Balances on-premise infrastructure with cloud services for flexibility and cost-efficiency.

9 Economized coverage Budget

9.1 Modifications to reduce costs

To optimize costs while maintaining performance and security, here's a list of specific services to modify and the potential savings for each adjustment.

- **Move Institutional Email to a Cloud Provider:** Modify: Instead of maintaining on-premise email servers, migrate to Google Workspace for Education (free for universities) or Microsoft 365 Education (low-cost).

Savings: \$100,000+ per year (hardware, maintenance, security).

- **Reduce On-Premise Storage by Using Cloud Backup:** keep active and critical data on-premise but move long-term archives and backups to AWS S3 Glacier or Google Cloud Storage.

Savings: \$50,000 per year (reducing expensive NAS storage needs).

- **Replace Paid Software with Open-Source Alternatives** Replace Paid Software with Open-Source Alternatives

1. Firewall: Use pfSense instead of Cisco ASA.
2. Monitoring: Use Zabbix instead of SolarWinds.
3. Authentication: Use FreeIPA instead of Active Directory.
4. Savings: \$75,000+ per year (eliminating software license fees).

- **Delay Non-Essential Upgrades:** postpone the purchase of non-critical hardware (high-end analytics tools, extra redundant servers) until more funding is secured. The upgrades may be postponed are:

1. Reduce 2 servers.
2. Reduce AP's from 230 to 180
3. Downgrade cable category from cat7 to cat6A
4. Reduce 500TB on-premise storage to 250TB by moving long-term backups to AWS S3 Glacier

Savings: \$75,000 per year.

- Use energy-efficient cooling (hot/cold aisle, liquid cooling) instead of expensive CRAC units.

Savings: \$50,000-%100.000 per year.

ITEM	ORIGINAL COST	ESTIMATED SAVINGS	NEW COST
Institutional Email Service	100.000 USD	50.000 USD	50.000 USD
On-Premise Storage (500TB → 250TB)	150.000 USD	75.000 USD	75.000 USD
Networking Equipment (APs Reduced: 230 → 180,)	448.600 USD	75.000 USD	373.600 USD
Security Software & Firewalls (Open Source Alternatives)	150.000 USD	75.000 USD	75.000 USD
Cloud Services (AWS Optimization)	250.000 USD	50.000 USD	200.000 USD
Infrastructure (Energy-Efficient Cooling & Power Optimization)	650.000 USD	95.000 USD	555.000 USD
Professional Services (Delaying Non-Essential Upgrades)	500.000 USD	20.000 USD	480.000 USD
TOTAL	2'108,600 USD	440.000 USD	1.808.600 USD

Figure 11: Estimation of the costs for the economized data-set.

These cost optimizations prioritize efficiency without compromising the university’s critical infrastructure.

- Moving institutional email to Education or Microsoft 365 eliminates the need for costly on-premise servers while improving security, availability, and user experience.
- Reducing on-premise storage from 500TB to 250TB does not impact daily operations, as frequently used data remains on high-speed local storage, while long-term archives and backups are securely stored in AWS S3 Glacier, ensuring cost savings and disaster recovery capabilities.
- Optimizing networking equipment by reducing WiFi access points (230 to 180) does not degrade performance, thanks to WiFi 6 technology and strategic placement in high-density areas.
- Similarly, switching to open-source security solutions, such as pfSense firewalls, Zabbix monitoring, and FreeIPA authentication, maintains enterprise-level security while eliminating expensive licensing fees.
- Infrastructure improvements, such as energy-efficient cooling, aisle containment, and smart PDUs, lower operational costs while maintaining optimal temperature control in the data center.

Part VI

CONCLUSIONS

10 Conclusions

- This project demonstrates that a high-performance and scalable IT infrastructure for the university can be achieved without excessive costs. By migrating non-critical services to the cloud, optimizing storage, networking, and cooling, and using open-source solutions, the university can maintain efficiency, security, and reliability while reducing operational expenses by approximately \$450,000.
- A well-designed IT infrastructure is the key to reducing operational costs while maintaining high performance, security, and scalability. Instead of relying on overprovisioned hardware and inefficient resource allocation, this project demonstrates that strategic planning and intelligent architecture can optimize expenses without sacrificing service quality.
- By integrating horizontal scaling, cloud optimization, and strategic cost reductions, the university's IT infrastructure remains highly performant, future-proof, and financially sustainable.
- A robust and well-structured network infrastructure directly enhances the quality of education by enabling fast, secure, and uninterrupted access to academic resources, virtual learning, and research tools.
- Every modern technology we rely on—cloud computing, artificial intelligence, online education, and digital communication—is built upon the foundation of networks. Though often unnoticed, networks are the silent yet powerful enablers of global connectivity.

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